SEED X

A farmer’s harvest should not lose value before it reaches the buyer.

Better Farming Starts with Better Connections.

CodeStorm Hackathon · Agriculture, FoodTech & Rural Development

Created & Developed by Tejash Shastri and Shaurya



Live demo

The value gap is real

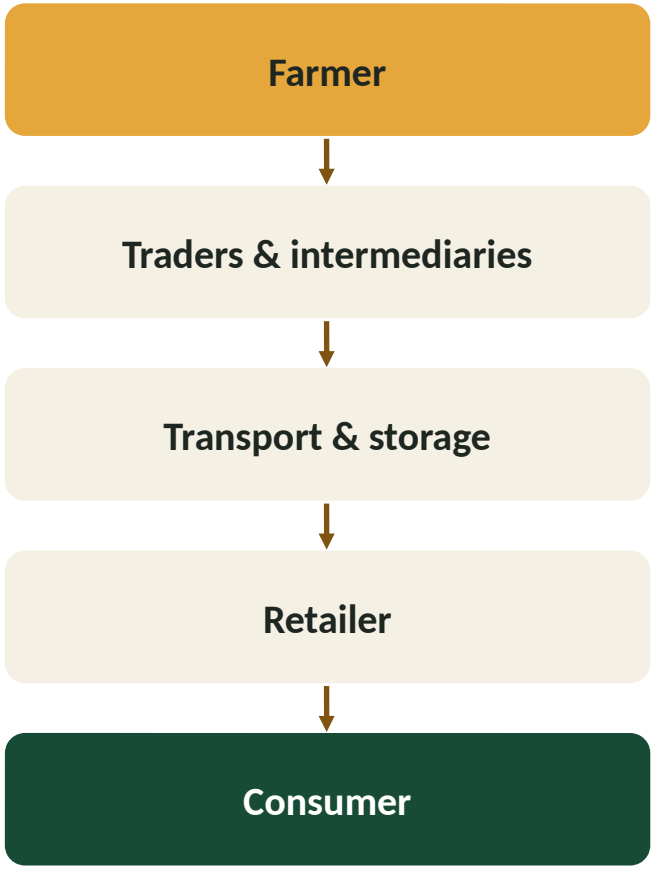
Estimated share of the consumer rupee that reaches the farmer

33%
Tomato

36%
Onion

37%
Potato

Roughly two-thirds of what a consumer pays for these vegetables is estimated to go beyond the farm gate.



Price rises at every hand-off

Source: RBI Working Paper 08/2024 (3 Oct 2024), "Vegetables Inflation in India: A Study of Tomato, Onion and Potato"; figures also cited in a Rajya Sabha answer (29 Nov 2024). Estimates from specific value-chain studies, not a universal figure for every crop, state or transaction.

ROOT CAUSE

The problem is not just selling. It is coordination

01

Too few buyers in view

Farmers may not see enough competing buyer demand.

02

Small lots are hard to find

Buyers struggle to find suitable local supply and small lots.

03

Fragmented pickups

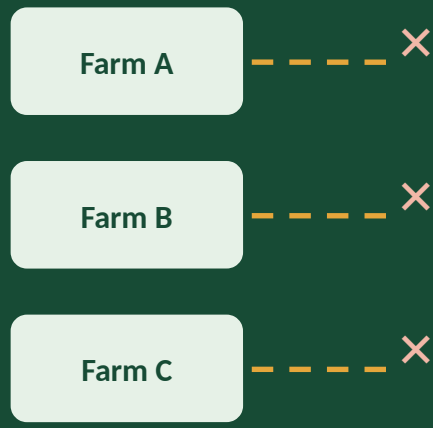
Separate trips for small orders can raise per-unit logistics cost.

04

Perishables lose value fast

Handling, transport and storage risks hit produce every day it waits.

Today



3 trips · 3 small sales · no shared view of demand

Context: Govt. of India / NABCONS post-harvest loss study (PIB, MoFPI, 1 Aug 2025) assessed losses across 54 crops at stages including collection, grading, packaging, transport and storage. SEED X is software; it does not build cold-chain infrastructure.

One software workflow from harvest to buyer

1 Farmer lists available harvest

2 Buyers post requirements or make offers

3 SEED X checks crop, grade, quantity, location and timing

4 ClusterCart compares pooled vs separate delivery

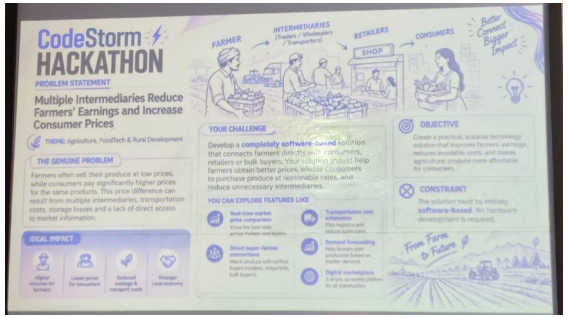
5 NetBack shows farmer net and buyer landed cost

6 Order, stock and delivery events tracked

Supporting services

- Crop & inventory
- My Clients / My Suppliers
- Notifications
- Seed store
- Mandi prices
- Govt. schemes
- 10 languages

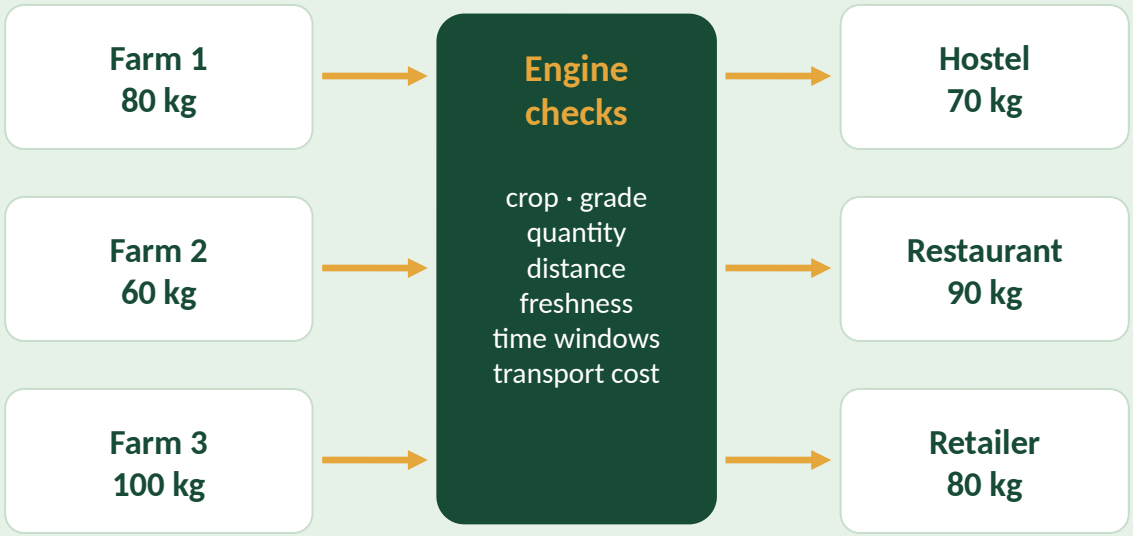
Software-only. No new hardware required.



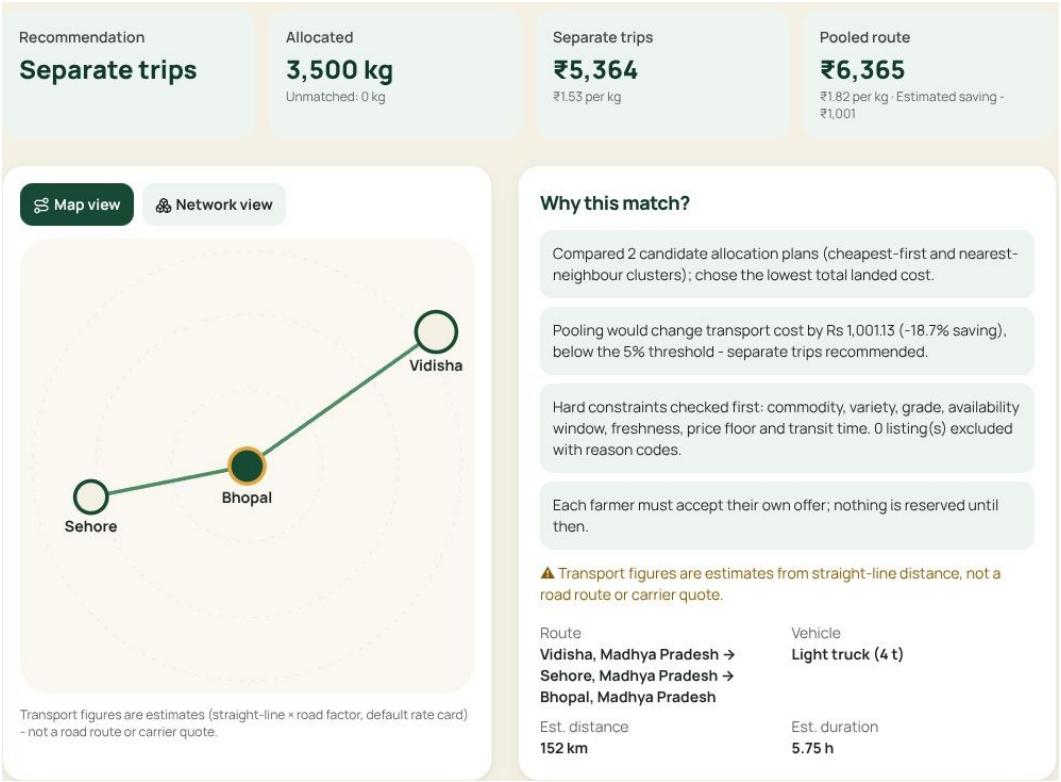
Official challenge: "Multiple Intermediaries Reduce Farmers' Earnings and Increase Consumer Prices"

Many farms. Many buyers. One smarter allocation

Illustrative scenario



240 kg supply = 240 kg demand. Pool only if the route is feasible and the estimated cost is lower; otherwise recommend separate trips.

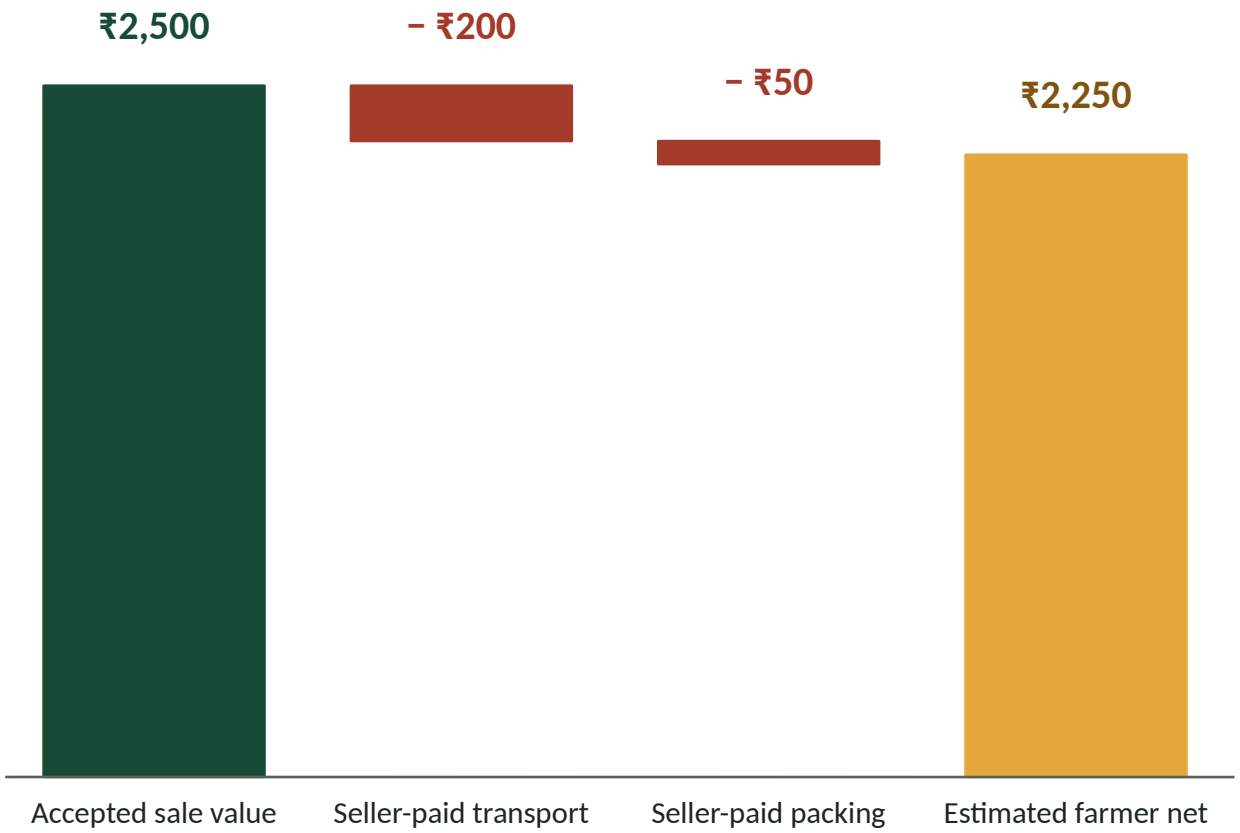


Live run on demo data: pooling would cost 18.7% more, so the engine said NO and recommended separate trips. It explains every decision.

Rule-based, explainable optimisation, not AI. Transport figures are estimates (straight-line distance × road factor, default rate card), not carrier quotes. Each farmer accepts their own offer.

A higher offer is not always a better deal

Illustrative example · farmer side

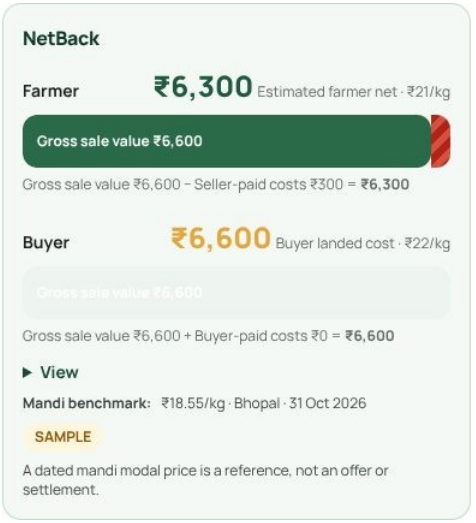


Buyer compares landed cost

Sale value
+ buyer-paid costs
= landed cost

Buyer-paid costs are never deducted from farmer net.

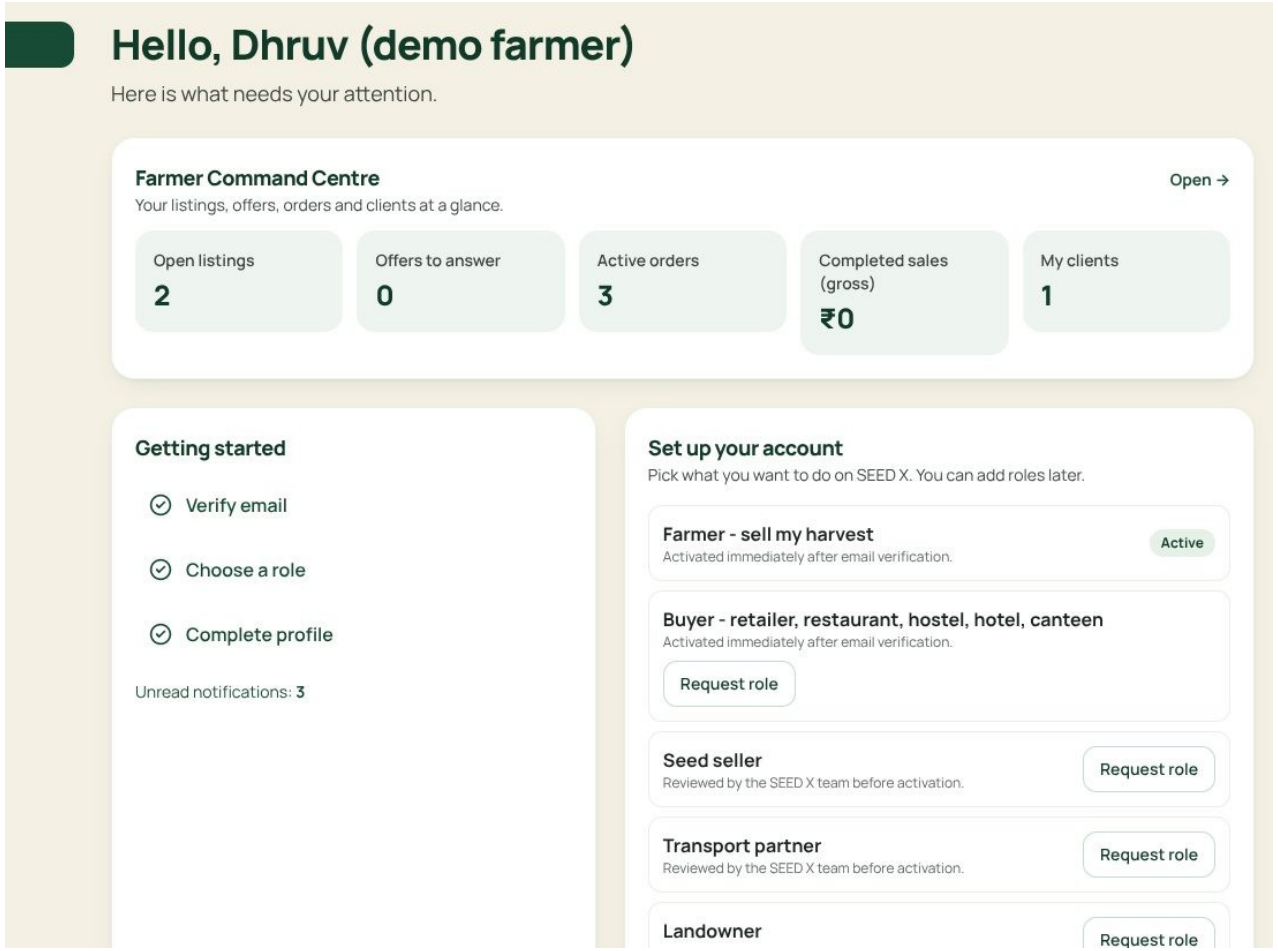
Farmers compare net proceeds. Buyers compare landed cost. Both see the same transparent breakdown.



Live app: 300 kg tomatoes at ₹22/kg, seller pays transport → est. farmer net ₹6,300; buyer landed ₹6,600 (demo data, estimated costs)

Illustrative numbers. Clearer offers and lower avoidable cost can create a path to better farmer outcomes and more affordable procurement; retail price pass-through is not guaranteed. Money is stored in whole paise.

A farmer’s daily business workspace



Manage

Harvest listings, 46-crop catalogue, available / reserved / sold stock that only changes through valid orders

Grow relationships

My Clients for farmers, My Suppliers for buyers: lead → enquiry → active → repeat, private notes

Complete trade

Offers, counter-offers, orders with timeline, logistics assignment, real-time notifications

Find support

Seed store with honest badges, dated mandi prices, official scheme links

All modules shown are working in the live app and covered by automated tests. Screenshot: demo farmer account (synthetic data).

Designed for the farmer, not for a tech expert

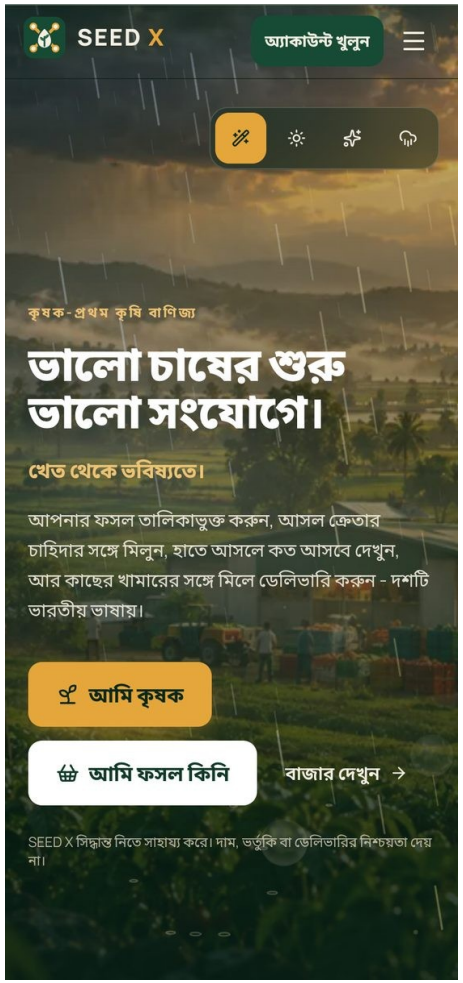
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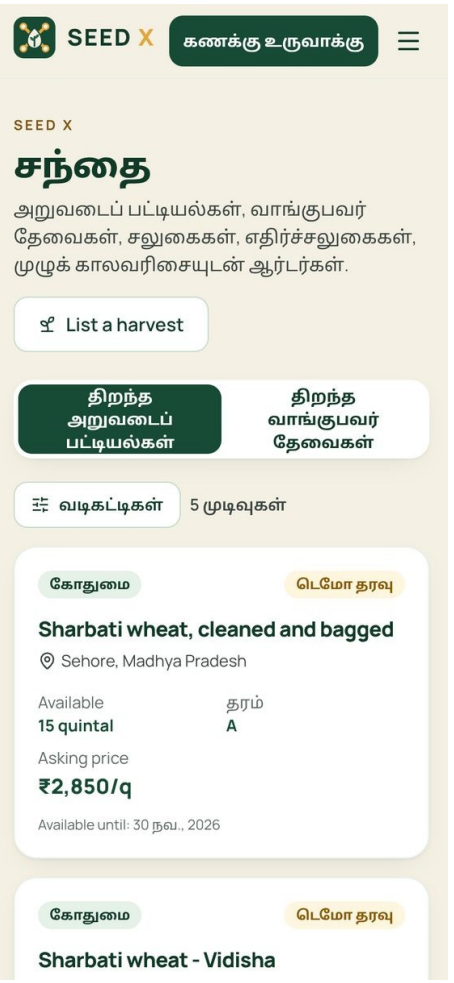
தமிழ் | ગુજરાતી | ಕನ್ನಡ | മലയാളം | ਪੰਜਾਬੀ

Ten languages live today in native scripts; core journeys fully translated, native-speaker review in progress.

- Mobile-first, large readable controls
- Short guided steps and clear kg / quintal units
- Local-language labels, helpful empty and error states
- Low-data design: works with 3D and animation off
- Assisted onboarding via FPOs / cooperatives as a rollout pathway (not a claimed partnership)

Validation plan: test core tasks with farmers in several languages; measure completion, errors and comprehension.





Context: IAMAI–Kantar "Internet in India 2024": 870 million users (98% of internet users) accessed Indic-language content in 2024. An all-internet-user statistic, not a farmer-specific survey. Screens: Bengali and Tamil, live app.

SECURITY & TRUST

Trust is built into the transaction

Verified accounts

Supabase Auth with email verification before any trading

Server-side roles

Roles are trusted records; admin actions need two-factor login (MFA)

Row Level Security

Enabled on all 40 tables: each user sees only their own private data

No secrets in the browser

Only the public publishable key ships; secret keys stay server-side

Atomic, idempotent orders

Stock reserved in one locked step; replays return the same order

Private files & audit

Owner-only upload folders, type/size limits, append-only audit log

Proof, not a shield icon

✓ DB security tests	141/141
✓ B reads A’s private notes	denied
✓ 2 buyers, last stock	1 winner
✓ Accept request replayed	same order
✓ User calls admin action	denied
✓ Upload to other’s folder	denied
✓ Browser + a11y tests	14/14

From the project’s automated pgTAP and Playwright suites, run against the live database on 31 Oct 2026. No system is "unhackable"; these are the specific controls that are configured and tested.

POSITIONING

Complement the ecosystem. Improve the daily workflow

Already exists

e-NAM

National electronic trading portal linking APMC mandis: price discovery, transparent auctions, payments, logistics module.

AGMARKNET / data.gov.in

Daily mandi minimum, maximum and modal prices.

Where SEED X focuses

- A local farmer workflow linking inventory, buyer demand, multi-party allocation, NetBack, relationships, orders and notifications
- Explainable pooled-delivery recommendation with a clear cost comparison, including when not to pool
- Mobile-first, ten-language workspace that uses public benchmark data where available

How it is built

Frontend

Vite · React 19 · TypeScript · Tailwind

Backend

Supabase Postgres · Auth · RLS · RPCs · Storage · Realtime · Edge Functions

Hosting

Cloudflare Workers, auto-deploy from GitHub

Visuals

Canvas 2D weather · Three.js for ClusterCart data only

Sources: enam.gov.in (overview, FAQs incl. logistics); data.gov.in / AGMARKNET current daily mandi prices. SEED X does not claim to be the first digital mandi; it is complementary to public market infrastructure. Entirely software-based.

DEMO · IMPACT · BUSINESS

Prove the journey in 60 seconds

1. Farmer’s tomato listing
2. Buyer requirement & offer
3. Run ClusterCart: see its checks
4. NetBack and buyer landed cost
5. Accept: stock, order and notification update

Live: seedxsss.tejash-shastri-rx.workers.dev
Offline backup: SEED_X_Demo_Backup.mp4 (79 s)

Pilot metrics we will measure

- Farmer net proceeds per kg vs a comparable alternative
- Buyer landed cost per kg
- Transport cost per kg
- Time to match
- Order fill and on-time delivery rate
- Repeat buyer rate
- Unsold / spoiled quantity

Proposed business model

Small fee per completed order

paid by buyers; free for farmers to list

ClusterCart planning

for institutional buyers (hostels, canteens, retailers)

Pilot first

one district, one crop cluster, partner FPO

Proposed, not yet validated

SEED X helps shorten the distance between a farmer’s harvest and a fairer market opportunity.

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